Homopolar generator through its ball bearings

Least drive torque per load watt when Tesla's two-disk homopolar takes its current through the shaft bearings — CLAUDE.md §3f, register entries H13, H14

kronspice simulation report · 2026-11-07 · 15 cm copper disks, 0.8 T; GA 702 evaluations per case

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Summary

Tesla's two-disk homopolar generator (U.S. Patent 406,968) is built with its output current taken through the shaft ball bearings instead of brushes, so that each bearing is also a Huber ball-bearing motor (the P element of *bearing_motor*). This is a circuit simulation only: every part is a lumped element and the shafts are mechanical nodes (across variable ω, through variable torque). The question is the least drive torque per watt delivered to an external resistor or capacitor. No coefficient is assumed; unmeasured ones are swept.

- **Reference unit** (40 mm bearings × 2 per shaft, 600 rad/s, 2 Ω): 41.80 W into the load from 77.29 W at the shaft, efficiency 0.541, 0.00308 N·m/W — but the bearing current is 7.62× the 0.3 A/mm² damage limit.
- **At the damage limit** a bearing bank passes only 0.238–1.248 A (bores 10–120 mm), so a through-bearing unit is a watts-to-tens-of-watts machine (2.22–9.50 W).
- **The Huber torque never helps the torque per watt:** in EM mode it lowers the drive torque only by lowering the output through its back-EMF; in thermal mode it leaves 41.797 W at 0.0031 N·m/W.
- **Insulated bearings do better:** 56.26 W at efficiency 0.661 with a shaft collector, against 41.80 W at 0.541 through the bearings.
- **Evolved:** through the bearings, 21.13 W at 0.0101 N·m/W (efficiency 0.134); bearings insulated, 689.28 W at 0.00135 N·m/W (efficiency 0.998), 7.4× less torque per watt.
- **Ledger:** residual 0 at every operating point; no free torque (shaft power is never below load plus measured losses).

Background

Register entries H13 and H14 concern rotating machines whose ledgers have been questioned: Tewari's space-power generator (H13), whose 24.8 N·m torque gap was compared with the bearing (Huber) torque, and Marinov's claim that a running ball-bearing motor yields heat beyond its electrical input (H14). The P element models the Huber effect from measured data only. This example asks a practical engineering question with the same element: if the homopolar current must pass through the bearings, what does it cost in drive torque? The standard prediction for every design is that the drive torque per load watt is at least 1/ω, with equality only at efficiency 1.

The trade-off and the measured inputs

The bearing bore sets the axle radius r_i, so bigger bearings remove magnetic area from each disk, EMF = ½Bω($r_o^2 - r_i^2$) (B: axial field, ω: shaft speed, r_o: disk radius). But bigger, more heavily loaded bearings carry more current before they are damaged, and have more friction.

input	value	source
bearing voltage under DC	almost constant, 0.73 V average at 4 A (0.5–1.5 V reported), from repeated breakdown of the oxide and lubricant film; carried by the P element as VF	Safdarzadeh, Farahi, Binder et al., Lubricants 12, 148 (2024); greased 51208 bearings, 40 mm bore
damage limit	0.1–0.3 A/mm² of Hertzian contact area; A_{Hz} = 1.0 mm² at 400 N and 2.5 mm² at 1600 N; other sizes by Hertz theory, A proportional to $(F \cdot R)^{2/3}$	same paper; Johnson, Contact Mechanics
friction torque	M = ½·f·F·d with f = 0.0015 (F: load, d: bore)	SKF simplified model

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input	value	source
rim-speed limit	hoop stress (3+ν)/8·ρv² at the yield of annealed copper, 69 MPa, safety factor 1.5: 111 m/s	Timoshenko; ASM Handbook
swept (unmeasured)	Huber coefficient κψ in EM and thermal modes; belt rolling-contact resistance; shaft collector resistance	—

TestHertzArea recovers the paper's second contact-area point from the first within 2 %.

Results (15 cm copper disks, 0.8 T)

Reference unit

Bore 40 mm × 2 per shaft, 400 N per bearing, disk r_o 0.150 m, r_i 0.020 m, thickness 5.0 mm, ω 600 rad/s (rim 90 m/s), load R_L 2 Ω. EMF 10.608 V; load 4.571 A, 41.80 W; shaft 77.29 W; efficiency 0.541; drive torque 0.0644 + 0.0644 N·m = 0.00308 N·m/W. Bearings: contact heat 6.70 W, friction 0.0240 N·m per shaft, Huber torque 0. The bearing current is 7.62× the 0.3 A/mm² limit (22.86× the 0.1 A/mm² one). Ledger residual 0.

Bearing bore against disk magnetic area (load matched to the bank's current limit)

bore	area kept	EMF	I_{max}	P_{load}	N·m/W	current / limit
10 mm	99.9 %	10.788 V	0.238 A	2.22 W	0.0073	1.00
20 mm	99.6 %	10.752 V	0.378 A	3.51 W	0.0088	1.00
40 mm	98.2 %	10.608 V	0.600 A	5.49 W	0.0107	1.00
60 mm	96.0 %	10.368 V	0.786 A	7.00 W	0.0122	1.00
80 mm	92.9 %	10.032 V	0.952 A	8.16 W	0.0137	1.00
100 mm	88.9 %	9.600 V	1.105 A	8.99 W	0.0153	1.00
120 mm	84.0 %	9.072 V	1.248 A	9.50 W	0.0171	1.00

Huber torque, κψ swept (reference unit)

κψ (N·m/A²)	mode	Huber torque	drive torque	P_{load}	N·m/W
0	EM / thermal	0.0000 N·m	0.1288 N·m	41.797 W	0.0031
10^{-3}	EM	0.0062 N·m	0.0978 N·m	24.738 W	0.0040
10^{-3}	thermal	0.0005 N·m	0.1278 N·m	41.797 W	0.0031
10^{-2}	EM	0.0065 N·m	0.0551 N·m	2.614 W	0.0211
10^{-2}	thermal	0.0006 N·m	0.1277 N·m	41.797 W	0.0031
10^{-1}	EM	0.0011 N·m	0.0484 N·m	0.044 W	1.1124
10^{-1}	thermal	0.0006 N·m	0.1277 N·m	41.797 W	0.0031

In EM mode the Huber torque lowers the drive torque only by lowering the output through its back-EMF, so it never improves the torque per watt. In thermal mode it recovers at most η·H, about 10 % of the bearing heat. *TestNoFreeTorque* checks that the shaft power is never below the load plus the measured losses.

Bearing-voltage range and the insulated alternative

configuration	P_{load}	efficiency	N·m/W
through bearings, 0.50 V each	46.11 W	0.578	0.0029
through bearings, 0.73 V each	41.80 W	0.541	0.0031
through bearings, 1.50 V each	28.91 W	0.418	0.0040
insulated bearings, collector 10^{-5} Ω	56.26 W	0.661	0.0025
insulated bearings, collector 10^{-4} Ω	56.25 W	0.661	0.0025
insulated bearings, collector 10^{-3} Ω	56.15 W	0.661	0.0025

Capacitor load: 1 F through R_L

The field is stepped (1 ms) or ramped, then the circuit settles for $5 \cdot R_L \cdot C$.

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field ramp	stored	shaft	friction	conversion	mismatch
0 s (stepped)	41.28 J	384.36 J	288.00 J	42.8 %	-1.6×10^{-3} J
20 s	41.80 J	927.73 J	864.00 J	65.6 %	7.3×10^{-5} J
100 s	41.96 J	3225.22 J	3168.00 J	73.3 %	7.0×10^{-5} J

The bearing drops cost $2\cdot V_b\cdot Q$ whatever the ramp (V_b: voltage per bearing, Q: charge delivered), so electrical conversion is at best $\tfrac{1}{2}CV^2/(\tfrac{1}{2}CV^2 + 2V_bCV) = 75.8$ % (V = 9.15 V), while friction runs throughout.

Evolved for the least drive torque per watt at ≥ 20 W

Genes: bore, preload, disk thickness, speed up to the rim limit, load and bearings per shaft; limits: the bearing current limit and the rim speed; 702 evaluations each.

	through the bearings	bearings insulated (collector 5×10^{-5} Ω declared)
design	108 mm × 6 per shaft, 185 N each, r_i 0.054 m, t 4.4 mm	10 mm × 1 per shaft, 50 N each, r_i 0.005 m, t 6.4 mm
speed; load	739 rad/s (rim 111 m/s); 4.84 Ω	739 rad/s (rim 111 m/s); 0.256 Ω
EMF	11.575 V	13.292 V
load	21.13 W at 2.089 A (1.00× the current limit)	689.28 W at 51.897 A
shaft power; efficiency	157.62 W; 0.134	690.38 W; 0.998
drive torque per watt	**0.0101 N·m/W**	**0.00135 N·m/W (7.4× less)**
ledger residual	0	0

The insulated unit's torque per watt is $1/(\eta\cdot\omega)$; in every case the drive torque per load watt is at least $1/\omega$.

Ledger and validation

- Ledger residual 0 at every operating point; transient mismatch below 10^{-4} of the shaft energy.
- *main_test.go*: Hertz scaling (*TestHertzArea*), no free torque (*TestNoFreeTorque*), bore monotonicity, the capacitor conversion ceiling.
- *export_test.go*: reference and capacitor netlists, KiCad schematics and results sheets in *export/*.

Limits and bench data still needed

- Circuit simulation only: lumped elements, shafts as mechanical nodes.
- The Huber coefficient κψ, the belt's rolling-contact resistance and the real collector's contact drop are unmeasured and swept. A carbon brush drops about 1 V, which would narrow the insulated design's lead; a liquid-metal collector would not.
- The damage limit is given as a range (0.1–0.3 A/mm²); the designs use 0.3 A/mm².

Reproducing the results

```
go run ./examples/homopolar_bearing [-p 20] [-gens 25] [-pop 30]
go test ./examples/homopolar_bearing
python examples/homopolar_bearing/viz/make_report.py examples/homopolar_bearing/homopolar_bearing_report.pdf
```

Sources: O. Safdarzadeh, A. Farahi, A. Binder et al., Lubricants 12(5), 148 (2024), DOI 10.3390/lubricants12050148; K. L. Johnson, Contact Mechanics, Cambridge 1985, §4.2; T. A. Harris, Rolling Bearing Analysis; SKF rolling-bearing catalogue; S. Timoshenko, Strength of Materials II, §44; ASM Handbook vol. 2 (C10100 annealed); N. Tesla, U.S. Patent 406,968 (1889); doc/bearing-motor.md.

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